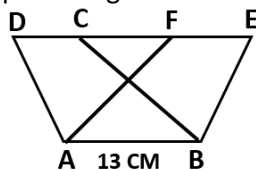




1. In the figure given below, if the area of parallelogram ABCD is 208cm^2 , what is the height of the parallelogram ABEF?



(Mensuration) NIMCET-2018

- (a) 15cm (b) 15.5cm
(c) 16cm (d) 16.5cm

2. The number of solid spheres, each of diameter 3 cm that could be moulded to form a solid metal cylinder of height 54 cm and diameter 4 cm is

(Mensuration) NIMCET-2019

- (a) 16 (b) 24
(c) 36 (d) 48

3. In a shower, 5cm of rain falls. The volume of water that falls on 1.5 hectares of ground is (Mensuration) NIMCET-2020

- (a) 75m^3 (b) 750m^3
(c) 7500m^3 (d) 75000m^3

ANSWER

1. (c)

2. (d)

3. (b)

Solution:

1. (c) Area of parallelogram ABCD = 208cm^2

$\therefore$ Area of parallelogram = Base $\times$ Height

Let height of parallelogram ABCD be h.

$$\therefore h \times 13 = 208$$

$$\therefore h = 16\text{ cm}$$

Now, height of parallelogram ABEF = Height of parallelogram ABCD

$\therefore$ Required height = 16 cm

2. (d) Given, diameter of solid sphere = 3cm

$$\therefore \text{Radius of solid sphere} = \frac{3}{2}\text{ cm}$$

$$\text{Solid sphere volume } (V_1) = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times \frac{3}{2} \times \frac{3}{2} \times \frac{3}{2}$$

$$= \frac{99}{7} \text{ cm}^3$$

Also given, height of the cylinder = 54 cm

and diameter of the cylinder = 4cm

$$\text{Radius of the cylinder} = \frac{4}{2} = 2\text{ cm}$$

$$\text{Volume of the cylinder } (V_2) = \pi R^2 h$$

$$= \frac{22}{7} \times 2 \times 2 \times 54$$

$$= \frac{4752}{7} \text{ cm}^3$$

Hence, the number of spheres

$$= \frac{V_2}{V_1}$$

$$= \frac{\frac{4752}{7}}{\frac{99}{7}} = \frac{4752}{99} = 48$$

3. (b) 1 hectare = 10000m^2
So, Area = $(15 \times 10000) \text{ m}^2$

